

A Compilation of Topics in Group Theory and Complex Analysis

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August 6, 2025

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1 Group Homomorphisms and Endomorphisms

1.1 Group Morphism (Homomorphism)

Definition 1.1 (Group Morphism). A **group morphism**, more commonly known as a **group homomorphism**, is a function between two groups that preserves the group operation. Formally, if $(G, *)$ and $(H, \cdot)$ are two groups, a function $\phi : G \rightarrow H$ is a group homomorphism if for all elements $a, b \in G$, the following holds:

$$\phi(a * b) = \phi(a) \cdot \phi(b)$$

where $*$ is the group operation in G and $\cdot$ is the group operation in H .

1.2 Endomorphisms

Definition 1.2 (Endomorphism). An **endomorphism** is a group homomorphism from a group to itself. If G is a group, an endomorphism of G is a function $\phi : G \rightarrow G$ that satisfies the homomorphism property.

1.2.1 Key Properties of Endomorphisms

Property 1.1 (Identity Preservation). An endomorphism always maps the identity element to itself: $\phi(e_G) = e_G$.

Property 1.2 (Ring Structure). The set of all endomorphisms of an abelian group G , denoted $\text{End}(G)$, forms a ring under the operations of pointwise addition and function composition.

1.2.2 Examples of Endomorphisms

Example 1.1 (Multiplication Map on Integers). Consider the group $(\mathbb{Z}, +)$ of integers under addition. For any integer n , the function $\phi_n : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $\phi_n(x) = nx$ is an endomorphism. This is because:

$$\phi_n(x + y) = n(x + y) = nx + ny = \phi_n(x) + \phi_n(y)$$

Example 1.2 (Frobenius Endomorphism). For an elliptic curve E defined over a finite field $\mathbb{F}_q$ where $q = p^k$, the **Frobenius endomorphism** $\phi : E \rightarrow E$ is defined by:

$$\phi(x, y) = (x^q, y^q)$$

This is a crucial endomorphism in elliptic curve cryptography.

2 The Concept of Degree for Homomorphisms

2.1 Degree in General Group Theory

In general abstract algebra, the term "degree" is not a standard property of a group homomorphism. While the size of the kernel is a fundamental concept (related to the First Isomorphism Theorem), it is not referred to as the "degree." The notion of degree is specific to certain contexts, such as morphisms between algebraic varieties.

2.2 Degree of an Isogeny on an Elliptic Curve

In the context of elliptic curves, an endomorphism is a special type of **isogeny**. For an isogeny, the degree is well-defined.

Definition 2.1 (Degree of a Separable Isogeny). *For a **separable isogeny** $\phi : E_1 \rightarrow E_2$ between two elliptic curves, the degree of ϕ , denoted $\deg(\phi)$, is defined as the size of its kernel:*

$$\deg(\phi) = |\ker(\phi)|$$

The kernel is the set of points on the source curve E_1 that are mapped to the identity element $\mathcal{O}$ on the target curve E_2 .

Example 2.1 (Degree of the Multiplication-by- m Map). *Consider the endomorphism $[m] : E \rightarrow E$ on an elliptic curve E , defined by $[m]P = mP$ for any point $P \in E$.*

- *The kernel of this map is the set of m -torsion points, $E[m] = \{P \in E \mid mP = \mathcal{O}\}$.*
- *Over a field whose characteristic does not divide m , the structure of the m -torsion subgroup is $E[m] \cong (\mathbb{Z}/m\mathbb{Z}) \oplus (\mathbb{Z}/m\mathbb{Z})$.*
- *Therefore, the size of the kernel is $|E[m]| = m^2$.*
- *Consequently, the degree of the isogeny $[m]$ is:*

$$\deg([m]) = m^2$$

This illustrates that the degree is not the scalar m but rather the size of the kernel, which is m^2 .

3 Isogeny vs. Homothety

3.1 Isogeny

Definition 3.1 (Isogeny). *An **isogeny** is a surjective morphism between abelian varieties (such as elliptic curves) that has a finite kernel. It is a structure-preserving map that respects the algebraic group structure.*

3.2 Homothety

Definition 3.2 (Homothety). *A **homothety** (or dilation) is a geometric transformation of an affine space that scales all vectors from a fixed center point S by a constant factor k . A point M is mapped to M' such that:*

$$S\vec{M}' = k \cdot S\vec{M}$$

Homotheties preserve angles and ratios of lengths but not necessarily lengths themselves.

Aspect	Isogeny	Homothety
Context	Algebraic geometry, group theory (elliptic curves, abelian varieties).	Euclidean/affine geometry, linear algebra.
Definition	A surjective group homomorphism with a finite kernel.	A geometric scaling transformation from a center point.
Structure	Preserves the algebraic group structure.	Preserves angles and similarity; an affine transformation.
Kernel	A finite subgroup (essential to the definition).	Has one fixed point (the center); not a group kernel.
Degree	For separable maps, the size of the kernel.	The scaling factor is its defining characteristic, not a "degree".

Table 1: Comparison of Isogeny and Homothety

3.3 Comparison

4 Tori in Lattices and Fields

4.1 Torus from a Lattice

Definition 4.1 (Torus from a Lattice). *In geometry and complex analysis, a **torus** can be constructed as the quotient space of a vector space by a lattice. For a lattice $\Lambda \subset \mathbb{R}^n$ generated by n linearly independent vectors, the torus is the quotient group:*

$$T = \mathbb{R}^n / \Lambda$$

This identifies any two points in $\mathbb{R}^n$ if they differ by a vector in the lattice Λ .

- In two dimensions, this space $\mathbb{R}^2/\mathbb{Z}^2$ is topologically equivalent to the familiar doughnut shape, $S^1 \times S^1$.
- In complex analysis, a **complex torus** $\mathbb{C}/\Lambda$ (where Λ is a 2-dimensional lattice in $\mathbb{C}$) is analytically isomorphic to an elliptic curve.

4.2 Prime Fields as Discrete Tori

The additive group of a prime field provides a finite analogue to the 1-dimensional torus.

- A 1-dimensional torus is the circle group S^1 , which can be represented as $\mathbb{R}/\mathbb{Z}$. This is a continuous group where arithmetic "wraps around" at every integer.
- The additive group of a prime field, $(\mathbb{F}_p, +)$, consists of the set $\{0, 1, \dots, p-1\}$ with addition modulo p . This group is isomorphic to $\mathbb{Z}/p\mathbb{Z}$.
- The "wrap-around" behavior, where $(p-1) + 1 = 0$, is analogous to the behavior on a circle. Thus, $(\mathbb{F}_p, +)$ can be viewed as a **discrete, finite analogue of a 1-dimensional torus**.

5 Laurent Series in Complex Analysis

5.1 Definition

Definition 5.1 (Laurent Series). A **Laurent series** is a representation of a complex function $f(z)$ as a power series that includes terms with negative powers. If $f(z)$ is analytic in an annulus $A = \{z \in \mathbb{C} : r < |z - z_0| < R\}$, its Laurent series expansion around z_0 is:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}$$

The first part is the **regular part**, and the second part is the **principal part**. The coefficients are given by the integral formula:

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{n+1}} dw$$

where C is a closed contour within the annulus A .

Laurent series are essential for analyzing functions near isolated singularities, where a Taylor series expansion would not be valid.

5.2 Example

Example 5.1. Let's find the Laurent series for $f(z) = \frac{e^z}{z^3}$ around $z_0 = 0$. The Taylor series for e^z is $e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$. Dividing by z^3 , we get:

$$f(z) = \frac{1}{z^3} \left(1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \dots \right) = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{2z} + \frac{1}{6} + \dots$$

The principal part is $\frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{2z}$.

5.3 Comparison with Taylor Series

Feature	Laurent Series	Taylor Series
Power Terms	Includes both non-negative and negative powers.	Includes only non-negative powers.
Region of Conv.	Converges in an annulus, which may contain a singularity at its center.	Converges in a disk where the function is analytic.
Applicability	Used for functions with isolated singularities.	Used for functions that are analytic at the center of expansion.

Table 2: Comparison of Laurent and Taylor Series

6 Endomorphisms and Isogenies of Elliptic Curves

An **endomorphism** of an elliptic curve E defined over a field K is a structure-preserving morphism from E to itself:

$$\varphi : E \rightarrow E,$$

which respects the group structure on E . Endomorphisms include:

- The identity map $\text{id} : P \mapsto P$,
- The multiplication-by- n map $[n] : P \mapsto nP$,
- The Frobenius endomorphism in the finite field setting,
- Other algebraic self-maps preserving the elliptic curve structure.

Endomorphisms form a ring, denoted by $\text{End}(E)$, under pointwise addition and composition.

An **isogeny** is a non-constant surjective morphism between two elliptic curves E_1 and E_2 (over the same field K),

$$\varphi : E_1 \rightarrow E_2,$$

which is a group homomorphism and takes the identity element of E_1 to that of E_2 . The kernel of an isogeny is a finite subgroup of E_1 . If $E_1 = E_2$, then an isogeny is an endomorphism.

Summary:

	Isogeny	Endomorphism
Definition	Morphism between two elliptic curves	Morphism from curve to itself
Domain, Codomain	$E_1 \rightarrow E_2$ (possibly different)	$E \rightarrow E$ (same curve)
Structure	Group homomorphism with finite kernel	Same, with ring structure
Kernel	Finite subgroup of domain curve	Finite subgroup (nonzero cases)
Examples	Vélu isogenies, Frobenius between curves	Identity, multiplication-by- n , Fro

7 Frobenius Endomorphism

In the setting of elliptic curves over a finite field $\mathbb{F}_q$ with $q = p^n$, the **Frobenius endomorphism** φ_q acts on points $P = (x, y) \in E(\mathbb{F}_q)$ by raising coordinates to the q -th power:

$$\varphi_q : (x, y) \mapsto (x^q, y^q).$$

- If P has coordinates in the base field $\mathbb{F}_q$ itself, then $\varphi_q(P) = P$ because for any $a \in \mathbb{F}_q$, $a^q = a$. Thus, the Frobenius endomorphism acts **trivially** on points defined over $\mathbb{F}_q$.
- For points over extension fields $\mathbb{F}_{q^m}$ with $m > 1$, φ_q acts as a nontrivial field automorphism, generally permuting the points:

$$\varphi_q(P) \neq P.$$

This endomorphism is fundamental in counting points on elliptic curves over finite fields and understanding their arithmetic structure.

8 Irreducible Polynomials and Elliptic Curves

Elliptic curves can be described algebraically by polynomial equations. The **short Weierstrass equation** over a field k is:

$$y^2 = x^3 + Ax + B,$$

where $A, B \in k$, with the **discriminant**

$$\Delta = -16(4A^3 + 27B^2) \neq 0,$$

to ensure the curve is nonsingular.

The polynomial

$$f(x, y) = y^2 - (x^3 + Ax + B)$$

is an **irreducible polynomial** over k when $\Delta \neq 0$. This irreducibility means that $f(x, y)$ cannot be factored into polynomials of lower degree with coefficients in the field k . This property is essential for the algebraic and geometric structure of the elliptic curve.

Summary:

- Irreducibility ensures the equation defines a smooth elliptic curve.
- Irreducible polynomials can be univariate or multivariate; here we consider the bivariate polynomial $f(x, y)$.

9 Irreducibility and Endomorphisms

An **endomorphism** $\alpha : E \rightarrow E$ of an elliptic curve E defined over a field K is an algebraic self-map preserving the group law. Such an endomorphism can be studied through its algebraic properties in the *endomorphism ring* $\text{End}(E)$.

9.1 Degree and Kernel of an Endomorphism

For a non-zero separable endomorphism α , the **degree** of α , denoted $\deg(\alpha)$, corresponds exactly to the size of its kernel subgroup:

$$\deg(\alpha) = |\ker(\alpha)|.$$

The degree is a measure of the “size” of the morphism as a finite map between varieties.

9.2 Irreducible Polynomial of an Endomorphism

Endomorphisms can often be identified with algebraic integers in a number field (for example, in the case of Complex Multiplication (CM)). Each such endomorphism α satisfies a **minimal polynomial** $m_\alpha(x)$ over the integers $\mathbb{Z}$:

$$m_\alpha(x) = x^2 - tx + d,$$

where $t, d \in \mathbb{Z}$, and m_α is irreducible over $\mathbb{Z}$ (or over $\mathbb{Q}$ if viewed as a polynomial with rational coefficients).

The **irreducibility** of $m_\alpha(x)$ ensures that α is an algebraic integer that cannot be decomposed into simpler algebraic elements.

9.3 Relation Between Degree, Kernel Size, and Irreducibility

The **degree** of the endomorphism α is related to the **norm** of the algebraic integer corresponding to α , which can be computed from its minimal polynomial:

$$\deg(\alpha) = \text{Norm}(\alpha) = d.$$

This norm equals the size of the kernel $\ker(\alpha)$, and d is typically the constant term in the irreducible minimal polynomial $m_\alpha(x)$. Hence:

$$\deg(\alpha) = |\ker(\alpha)| = d,$$

where d arises from